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DOCUMENTATION

Pizza Sales Analysis and Prediction: Identifying Factors Influencing Best and Worst Selling Pizzas

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ABSTRACT:

The project “Pizza Sales Analysis and Prediction: Identifying Factors Influencing Best and Worst Selling Pizzas” focuses on performing Exploratory Data Analysis (EDA) on a pizza restaurant’s sales dataset to understand sales trends and identify factors affecting pizza performance. The dataset, obtained from transactional sales records, includes attributes such as pizza name, category, size, quantity sold, unit price, total sales, and ingredients.

Using Python-based data analysis tools such as Pandas, Matplotlib, Seaborn, and Scikit-learn, the data was cleaned, summarized, and visualized to explore relationships among variables. EDA techniques like grouping, aggregation, and correlation analysis were applied to identify the best and worst-selling pizzas and to examine how factors like price, size, and category influence overall sales. A simple Linear Regression model was used to study the impact of these features on sales quantity, while ingredient frequency analysis helped in understanding common preferences among top-selling pizzas.

Through this analysis, meaningful insights were derived to help improve business performance, including adjusting pizza sizes and prices, enhancing less popular items with common ingredients, and using promotional offers to increase sales. The project demonstrates how basic EDA methods and regression analysis can transform raw restaurant data into actionable insights for better decision-making.

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INTRODUCTION

The modern restaurant industry operates in a dynamic and highly competitive environment where customer preferences, menu diversity, and pricing strategies directly influence profitability. To remain competitive, restaurants must adopt a data-driven approach rather than relying solely on experience or intuition. By analyzing historical sales data, businesses can uncover valuable insights about consumer behavior, product performance, and pricing effectiveness, enabling them to make informed and strategic decisions.

This project applies Exploratory Data Analysis (EDA) techniques to a pizza restaurant's sales dataset to identify meaningful patterns and relationships among product attributes such as pizza name, category, size, price, and ingredients. The goal is to determine which pizzas perform best and which perform poorly, and to understand the underlying factors influencing their sales. Using Python's analytical libraries for data visualization and a simple Linear Regression model, the study explores how pricing and size affect overall demand and product performance.

The insights derived from this analysis help restaurant managers understand which product characteristics attract customers, which deter them, and how to improve overall sales performance. Additionally, the study examines ingredient popularity to suggest enhancements for underperforming pizzas. Ultimately, this project demonstrates how EDA combined with basic predictive analysis can guide real-world business decisions in the food industry by transforming raw sales data into actionable strategies for growth and profitability.

1.1 Problem Statement

In the restaurant industry, understanding what drives the success or failure of specific menu items is essential for maximizing sales and customer satisfaction. Many restaurants continue to make pricing, portioning, and promotional decisions based on assumptions or intuition rather than evidence-based insights. This often leads to inconsistent sales performance and missed opportunities for improvement. In the restaurant industry, understanding what drives the success or failure of specific menu items is essential for maximizing sales and customer satisfaction. Many restaurants continue to make pricing, portioning, and promotional decisions based on assumptions or intuition rather than evidence-based insights. This often leads to inconsistent sales performance and missed opportunities for improvement.

This project aims to answer the key question:

“Which pizzas sell the best and worst, and what factors contribute to their sales performance?”

By using data-driven insights, this study identifies attributes that lead to both high and low pizza sales and provides rule-based recommendations to improve poorly performing items. The goal is to help the restaurant make informed business decisions through the power of data analysis rather than assumptions.

1.2 Objective

The main objective of this project is to explore and analyze pizza sales data using Exploratory Data Analysis (EDA) techniques to uncover meaningful patterns that can assist in making informed business decisions. The project aims to identify which pizzas perform best, which underperform, and the factors contributing to their sales performance. By examining features such as price, category, size, and ingredients, the analysis provides insights into customer preferences and overall sales behavior.

The study utilizes Python's data analysis and visualization libraries — including Pandas, Matplotlib, Seaborn, and Scikit-learn — to process, analyze, and visualize the dataset. The purpose is not only to understand existing sales trends but also to derive actionable insights that can help optimize pricing, improve product composition, and enhance menu offerings.

The specific goals of the project are to:

- Perform data cleaning and preprocessing to ensure the dataset is accurate, consistent, and ready for analysis.
- Conduct exploratory data analysis (EDA) to uncover sales patterns, category performance, and customer preferences.
- Identify best and worst-selling pizzas based on total quantity sold and total revenue generated.
- Analyze how price, size, and category affect pizza sales using simple linear regression and visual exploration.
- Perform ingredient frequency analysis to determine which toppings are most common in top-selling pizzas.
- Provide business insights and recommendations to improve low-performing pizzas through pricing strategies, promotional offers, and ingredient optimization.

1.3 Significance of the Study

This project holds great significance in showcasing how data analytics can help restaurant businesses move beyond guesswork and make informed decisions. By analyzing pizza sales data, the study provides valuable insights into customer buying behavior, identifying which pizzas perform best and why certain variants fail to attract demand. Understanding how attributes such as price, size, and ingredients influence sales helps the restaurant make evidence-based improvements in menu design, pricing, and marketing strategies.

Additionally, this project demonstrates the practical application of Exploratory Data Analysis (EDA) and simple Linear Regression techniques to extract business intelligence from real-world data. The approach highlights how even basic analytical methods can uncover meaningful patterns and trends, guiding better operational and promotional decisions. The study ultimately contributes to improving overall sales performance and customer satisfaction through data-driven insights.

DATA COLLECTION AND PREPARATION

2.1 Data Source

The dataset used in this project was obtained from a pizza restaurant's internal sales database, containing detailed transactional records of pizza sales for the year 2015. It includes essential information such as pizza name, category, size, quantity sold, unit price, total sales, and ingredients. This dataset provides a realistic view of business operations and serves as an excellent foundation for understanding customer preferences, pricing effectiveness, and product performance across various pizza types.

The data was provided in Microsoft Excel format under the title "Data Model - Pizza Sales.xlsx" and was imported into Python using the Pandas library for analysis. Once imported, initial exploration was carried out using basic functions such as `.head()`, `.info()`, and `.isnull().sum()` to examine data types, structure, and completeness. This preliminary inspection helped identify the relevant columns required for analysis, validate data accuracy, and ensure consistency before performing exploratory data analysis (EDA).

Through this process, the dataset was prepared for further cleaning, aggregation, and feature engineering steps that enabled the identification of best- and worst-selling pizzas and the factors influencing their sales.

2.2 Data Description

The dataset contains multiple attributes describing each pizza sale transaction. Each record corresponds to an individual pizza item sold, along with its features such as size, price, and category.

The key attributes include:

- `order_id` – Unique identifier for each order placed by customers.
- `pizza_id` – Unique identifier for each pizza variant.
- `pizza_name` – The name of the pizza sold (e.g., “The Big Meat Pizza”, “The Greek Pizza”).
- `pizza_category` – Type or classification of the pizza (e.g., Classic, Supreme, Veggie, Chicken).
- `pizza_size` – Indicates pizza size: Small (S), Medium (M), Large (L), Extra Large (XL), or Double Extra Large (XXL).
- `quantity` – The number of units sold in a particular order.
- `unit_price` – Price of one pizza of the specified type and size.
- `total_sales` – Total sales value calculated as $quantity \times unit_price$.
- `pizza_ingredients` – Ingredients or toppings used in each pizza (used for flavor and ingredient frequency analysis).

The dataset includes both numerical attributes (`quantity`, `unit_price`, `total_sales`) and categorical attributes (`pizza_category`, `pizza_size`, `pizza_name`). This combination allows for both statistical and visual analysis to understand how various attributes influence pizza sales.

2.3 Data Preprocessing

Before performing Exploratory Data Analysis (EDA) and model training, several preprocessing steps were carried out to ensure the dataset's accuracy, consistency, and readiness for analysis.

1. Handling Missing Values:

Checked for null or missing entries using `.isnull().sum()` to ensure data completeness and reliability.

2. Feature Creation:

Created a new column, `total_sales`, by multiplying *quantity* and *unit_price*, ensuring that the calculated values matched the existing *total_price* column for consistency.

3. Data Aggregation:

Grouped the dataset by `pizza_name`, `pizza_category`, and `pizza_size` to calculate the total quantity sold, total revenue, and average unit price for each pizza. This helped summarize overall sales trends and identify best- and worst-performing pizzas efficiently.

4. Feature Encoding:

Categorical variables such as `pizza_category` and `pizza_size` were converted into numeric form using `LabelEncoder` from `Scikit-learn`, preparing the data for regression modeling.

5. Ingredient Frequency Processing:

The `pizza_ingredients` column, containing comma-separated ingredient lists, was split and transformed into individual elements using `.str.split()` and `.explode()`. Frequency counts were generated for each ingredient to identify the most common toppings. This process supported the ingredient frequency visualization and revealed which ingredients were most prevalent in top-selling pizzas.

6. Data Consistency Check:

Cross-verified that calculated totals, aggregated quantities, and encoded variables maintained consistency across all records, ensuring the dataset was clean and ready for analysis.

METHODOLOGY

The methodology outlines the systematic process followed to analyze the pizza sales dataset and extract meaningful business insights. This project adopts a data-driven analytical approach, combining data preprocessing, exploratory data analysis (EDA), visualization, and a simple regression model to study the relationship between pizza attributes and their sales performance.

The dataset used in this project, titled “Data Model – Pizza Sales.xlsx,” contains detailed transactional records from a pizza restaurant, including information such as pizza name, category, size, unit price, quantity sold, total revenue, and ingredients. These features were analyzed statistically and visually to uncover sales trends, customer preferences, and factors contributing to high or low product performance.

The overall process involved data loading and cleaning, feature aggregation, and pattern recognition through visualization. A Linear Regression model was applied to determine how pricing, size, and category influence sales volume, while ingredient frequency analysis was conducted to identify popular toppings among best-selling pizzas. This structured methodology ensured that insights were not only descriptive but also actionable for improving sales strategies and menu planning.

3.1 Approach

The entire analysis follows a structured and repeatable workflow designed to ensure accuracy, clarity, and reproducibility. The process began with a thorough understanding of the dataset, identifying relevant variables such as pizza name, category, size, price, and quantity sold. This was followed by data cleaning and preprocessing, where missing values were checked, and derived metrics like total sales were calculated to validate the data’s consistency.

After preparing the dataset, Exploratory Data Analysis (EDA) was conducted to uncover sales trends and relationships between features. The data was aggregated based on pizza attributes to compare performance, followed by visualizations that provided a clearer understanding of customer preferences and pricing impact.

A Linear Regression model was then applied to examine how variables like price, size, and category influenced sales quantity. Finally, ingredient frequency analysis was performed to identify commonly used toppings in best-selling pizzas, providing insights for flavor improvement and menu optimization.

3.2 Tools and Libraries Used

To perform data cleaning, manipulation, visualization, and analysis efficiently, several Python libraries and analytical tools were used. These tools provided a flexible environment for conducting Exploratory Data Analysis (EDA) and applying simple predictive modeling.

The following technologies were utilized in this project:

- Python: The core programming language used for data processing, visualization, and analysis due to its simplicity and wide range of data analysis libraries.
- Pandas: Used for data loading, cleaning, transformation, and aggregation. It provided powerful data structures like DataFrames for handling large datasets efficiently.
- Matplotlib: Employed for creating static visualizations such as bar charts and scatter plots to display sales trends and model comparisons.
- Seaborn: Built on top of Matplotlib, it was used to generate aesthetically appealing and informative statistical visualizations, including correlation heatmaps and feature importance charts.
- Scikit-learn: Utilized for encoding categorical variables using LabelEncoder and implementing the Linear Regression algorithm to identify relationships between pizza features and sales.
- Jupyter Notebook / Google Colab: Served as the interactive development environment for executing Python code, visualizing outputs, and documenting each step of the analysis process.

3.3 ALGORITHM USED (Linear Regression)

The Linear Regression algorithm was applied in this project to analyze how different pizza attributes, specifically unit price, pizza category, and pizza size influence the sales quantity of each pizza type. The model was chosen for its simplicity, interpretability, and effectiveness in understanding linear relationships between multiple independent variables and a continuous dependent variable.

Before training, the dataset underwent preprocessing, including encoding of categorical features using LabelEncoder and aggregating total sales for each pizza type. The features (unit_price, pizza_category_encoded, and pizza_size_encoded) were used as independent variables, while the target variable (quantity) represented the number of pizzas sold.

The Linear Regression model was trained using Scikit-learn's LinearRegression() function to quantify the relationship between sales and product attributes. The model achieved consistent results, showing that price and size have a negative effect on sales, confirming that larger and higher-priced pizzas sell less frequently. The regression coefficients also provided clear interpretability, allowing direct comparison of how strongly each factor impacts overall sales.

This algorithm was not used for prediction purposes but rather as an analytical tool to validate findings from the Exploratory Data Analysis (EDA). Its efficiency lies in highlighting the quantitative influence of key variables, supporting the data-driven recommendations made for improving sales performance.

RESEARCH AND ANALYSIS

4.1 Results

The analysis of the pizza sales dataset successfully revealed patterns in customer preferences and product performance. Through Exploratory Data Analysis (EDA), the data was grouped and visualized to identify which pizzas performed the best and which sold the least. The results showed that “The Big Meat Pizza (Small)” was the highest-selling pizza, while “The Greek Pizza (XXL)” recorded the lowest sales.

Further analysis of the attributes revealed that pizza size and price have a significant negative impact on sales quantity. Smaller, affordable pizzas were purchased more frequently than larger and higher-priced ones. The Linear Regression model, though used in a simple and supportive role, confirmed this relationship by showing that as price or size increases, sales quantity decreases.

Overall, the results clearly indicated that customer preference leans toward smaller, value-based pizzas with common ingredients. These findings align with the visual and statistical patterns observed during EDA.

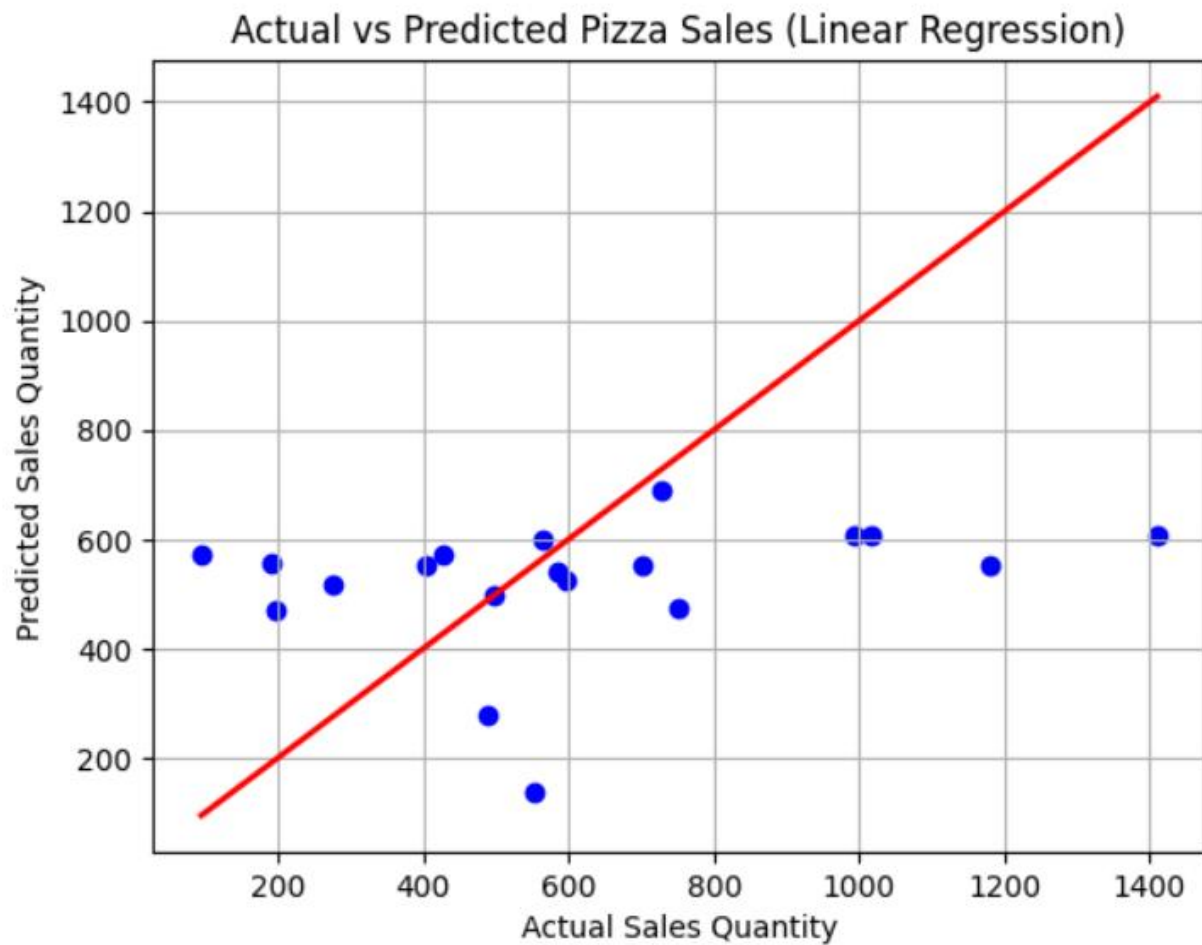
4.2 Visualization

Visualization played an important role in understanding sales patterns and feature relationships.

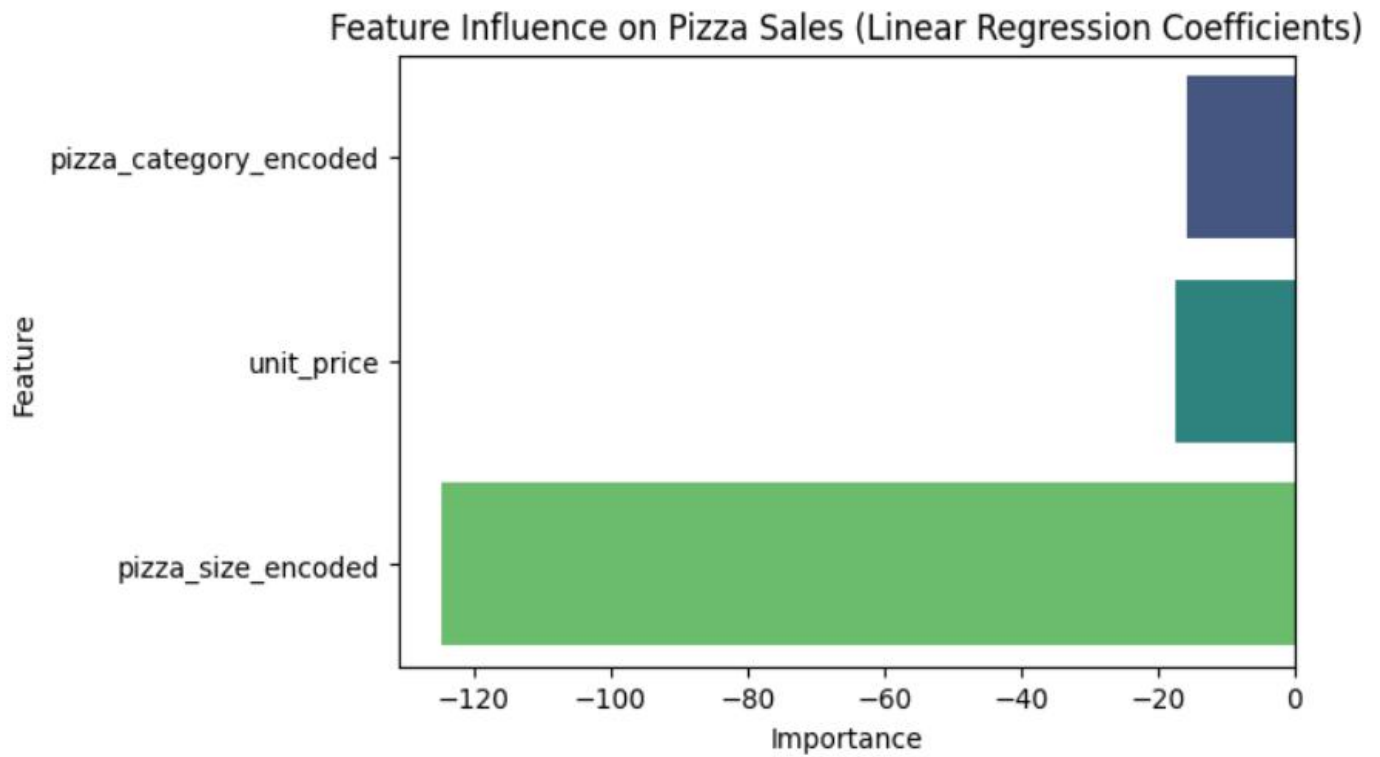
Using Matplotlib and Seaborn, key charts were created to interpret results clearly:

- Scatter Plot: Showed the relationship between actual and predicted sales, confirming that the Linear Regression model followed real sales trends.
- Feature Coefficient Chart: Highlighted how price, size, and category influenced sales — with price and size showing a negative impact on demand.
- Ingredient Frequency Chart: Displayed the top ingredients such as mozzarella, mushrooms, onions, tomatoes, and pepperoni, commonly found in best-selling pizzas.

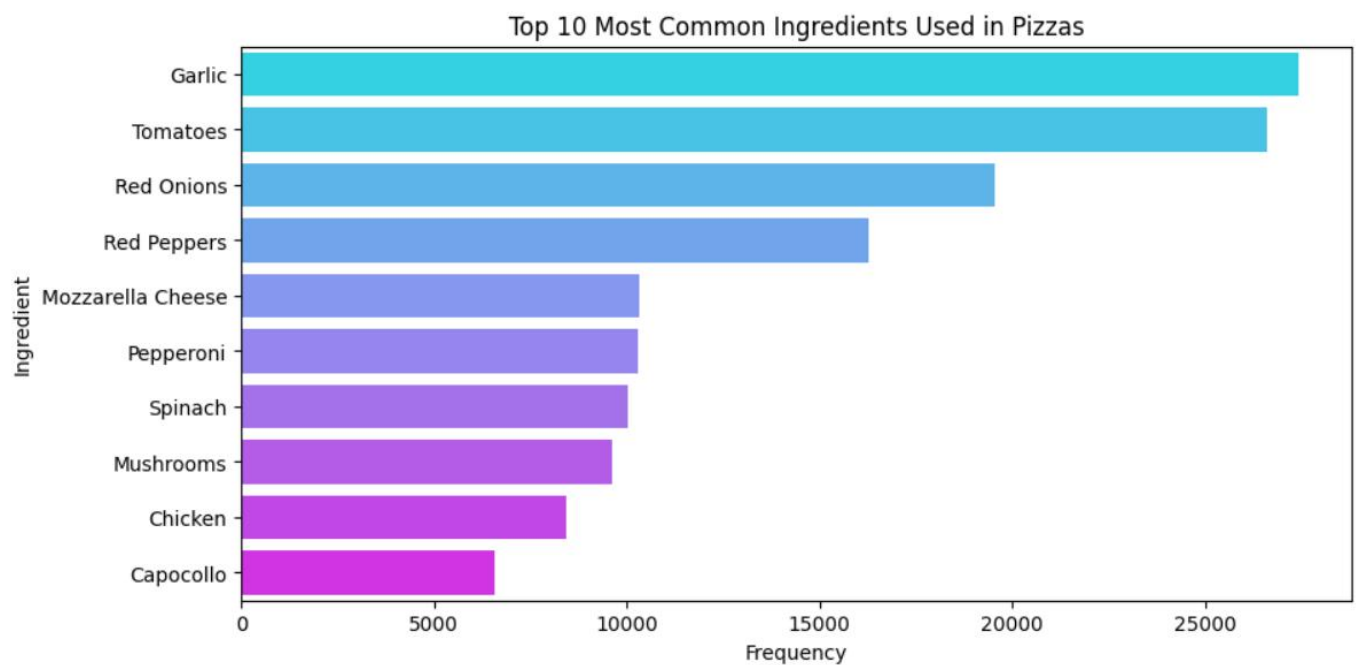
Scatter plot showing the relationship between actual and predicted pizza sales quantities.



Bar chart illustrating how features like price, size, and category influence pizza sales.



Horizontal bar chart displaying the top 10 most common ingredients used in pizzas.



4.3 Analysis

The project primarily focused on using Exploratory Data Analysis (EDA) to extract data-driven insights rather than implementing complex machine learning models. The Linear Regression model was applied as a simple yet effective analytical method to validate the relationships identified during the exploratory phase, particularly the impact of features such as price, size, and category on sales performance.

The analysis revealed that price and size were the most influential factors affecting pizza sales. Larger and higher-priced pizzas generally experienced lower demand, whereas smaller and mid-sized pizzas offered better value for money and achieved higher sales volumes. Ingredient analysis further showed that pizzas containing popular toppings like mozzarella, Red onion, Garlic and pepperoni tended to perform better, highlighting the strong link between ingredient appeal and customer preference.

From these findings, several business recommendations were derived. Restaurants can introduce limited-time discounts or promotional offers on large and premium pizzas to boost their demand. Offering combo deals that bundle large pizzas with beverages or side dishes can also make higher-priced items more appealing. Additionally, improving underperforming pizzas by adding widely preferred ingredients from top-selling varieties can help enhance flavor appeal and drive sales. Finally, better menu placement and targeted marketing strategies can increase visibility and customer interest in low-selling pizzas.

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project, “Pizza Sales Analysis and Prediction: Identifying Factors Influencing Best and Worst Selling Pizzas,” focused on understanding pizza sales patterns using Exploratory Data Analysis (EDA) and a simple Linear Regression model. The analysis revealed that factors such as price and size play a crucial role in determining sales performance. Smaller-sized and moderately priced pizzas were found to perform better in terms of quantity sold, while larger and costlier pizzas showed lower demand.

Additionally, through ingredient frequency analysis, it was observed that top-selling pizzas contained popular ingredients like mozzarella, mushrooms, and pepperoni, which appeal to a wide range of customers. Based on these findings, several strategies were suggested to improve the sales of poorly performing pizzas, including price adjustments, combo offers, and flavor enhancement using commonly preferred ingredients. This project demonstrates how simple analytical techniques can provide valuable business insights, even without the use of complex machine learning models.

5.2 Future Work

While this project provides valuable insights using basic EDA and Linear Regression, there is significant potential for further enhancement. Future improvements could include:

1. Incorporating More Features:

Including additional variables such as customer demographics, time of purchase, and location data could give deeper insights into sales patterns.

2. Advanced Predictive Models:

More complex machine learning models like Decision Trees, Random Forest, or K-Nearest Neighbors (KNN) can be implemented to improve prediction accuracy and uncover hidden relationships.

3. Seasonal and Time-Series Analysis:

Analyzing sales trends over time can help identify peak seasons, popular hours, or weekdays with higher sales, aiding inventory and marketing decisions.

4. Recommendation System:

A simple recommendation model can be developed to suggest ingredient combinations or menu modifications based on customer preferences and frequent itemsets.

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